

FIELDNODE · DESIGN DECISION RECORD

FieldNode recommendations accepted

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DOCUMENT CONTROL

Document ID	FND-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/fieldnode		
Source	docs/decisions/0002-recommendations-accepted.md @ bf605a8		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all recommendations and what changed in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below marked "Decided" is decided by Amish, 2026-09-25: go with recommendation. Items without a recommendation remain "Proposed, awaiting Amish".

■ Context

After the TRL 3 session, FieldNode had twelve items adopted for TRL 3 work pending Amish's review (FND-DDR-001, D1 to D12) and five new items in `docs/REVIEW.md` (session 2026-09-25, TRL 3, items 4 to 8), each with a recommendation. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Where a recommendation offered several options, the recommended option is the decision. Items that carried no recommendation are not decided by this record.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1 to D12	FND-DDR-001 items: LoRaWAN radio; STM32WL-class module; one 6 Ah LiFePO4 cell with its own 1S protection; 6 W panel standard; two M12 5-pin ports plus a gland; 15 min default interval; TwinKit first with The Things Network as fallback; stock polycarbonate IP65 enclosure; panel as hood; antenna down; 1.75 m mounting height; no change to budget, pitch or problem	As recommended in FND-DDR-001	FND-DDR-001 v0.2 status wording; FND-PRC-001 v0.4 "Key design choices" now decided. <code>budget_usd</code> stays \$150; pitch and problem unchanged
N1	Sun shield (REVIEW item 4)	Option (b): a ventilated white aluminium shield only at hot-climate sites; R14 applies to the base node	BOM line 14 added as an option at qty 0 (\$8.00); <code>build_shield()</code> in <code>cad/src/model.py</code> and <code>cad/step/fieldnode-shield.step</code> and <code>.stl</code> ; FND-CAL-001 v0.2 finds the threshold: the shield is fitted where the site's design maximum exceeds 30 °C; R3 restated; drawing note on FND-DWG-001 Rev P2
N2	R14 mass (REVIEW item 5)	Keep 2.5 kg for the base node	R14 restated to exclude the shield; base node 2.41 kg, hot-climate node 2.55 kg
N3	Panel voltage class (REVIEW item 6)	9 V class panel of the same power, subject to the charger chosen at TRL 4	BOM line 4 respecified at the same price; FND-CAL-001 [C6] now 7.42 V when hot, 2.42 V above a 5 V charger minimum input (was 4.95 V, 0.05 V short); drawing note, precis, README
N4	Published sensor allowance (REVIEW item 7)	100 mW	R7 notes 100 mW as published; precis, README, blueprint key figures; autonomy 5.75 days, 15 % margin; sibling notes citing 115 mW listed as a cross-repo action
N5	Firmware airtime rule (REVIEW item 8)	On The Things Network, lengthen the interval automatically at SF10 and slower: 22 min at SF10, 48 min at SF11, 87 min at SF12	R9 restated to include the rule; FND-CAL-001 [B1b] shows 30.0 s a day or less at every spreading factor; precis airtime table. Writing the firmware is TRL 4 work and on hold

Table 2. Items still open.

#	Item	Status
O1	First adopting projects and pilot region, which set the radio band and antenna. No recommendation was made.	Proposed, awaiting Amish
O2	Sensor port pinout, to be agreed with the first two adopting projects. FND-CAL-001 offers a candidate for discussion only, not a recommendation.	Proposed, awaiting Amish and the adopting project teams
O3	Firmware update method in the field (sealed USB port or over the air). No recommendation was made.	Proposed, awaiting Amish

The suggestion in `docs/REVIEW.md` of a panel-powered cell heater for long sub-zero spells was a suggestion, not a recommendation awaiting decision, and is not adopted.

■ Consequences

- Requirement status (FND-CAL-001 v0.2): none not met, 2 at risk (R2 cold charging, R6 cold or aged cell), 10 met on paper, 5 met by design, 1 not verifiable at TRL 3 (R12 install time). Before: 1 not met, 5 at risk, 6 met on paper.
- Controlled documents revised: FND-PRB-001 v0.4, FND-PRC-001 v0.4, FND-REQ-001 v0.4, FND-CAL-001 v0.2, FND-DDR-001 v0.2; drawing FND-DWG-001 Rev P2.
- Cost: base node \$126.00 (unchanged); hot-climate node \$134.00. Both within the \$150 `budget_usd`, which is unchanged.
- Mass: base node 2.41 kg; hot-climate node 2.55 kg.
- Cross-repo actions (other repos not edited): see `docs/REVIEW.md`, session "recommendations accepted".
- TRL 4 remains on hold by Amish's instruction. The charger part choice, the firmware rule's implementation and any test of the shield factor are TRL 4 work and have not been started.